Predicting Reaction Success

A Binary Classifier with k-Way Combinatorial Fusion Analysis

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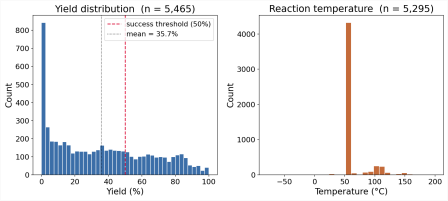
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Machine Learning Final Project

The Problem

Question: given the ingredients of a chemical reaction, will it produce a usable amount of product?

- Inputs: SMILES strings for the 6 reaction components + temperature.
- Output: **success** (yield $\geq 50\,\%$) or **failure**.
- Why 50 %: synthetic-chemistry literature's "preparatively useful" threshold; below it side reactions dominate.
- Use case: **bench-chemistry triage** — predict before running the experiment.



Why classification, not regression?

The first version of this project predicted continuous yield. On unseen chemistry it **collapsed**.

- Random split: regression $R^2 = 0.82$ (state-of-the-art).
- Held-out scaffolds: $R^2 < 0$ for **every** model tested.

Why this pivot makes the project more, not less, scientifically honest:

- Bench chemists make a binary decision (try / don't try); they *might not* need absolute yield precision.
- Ranking-based ensembling (CFA) operates naturally on a binary task.

Dataset

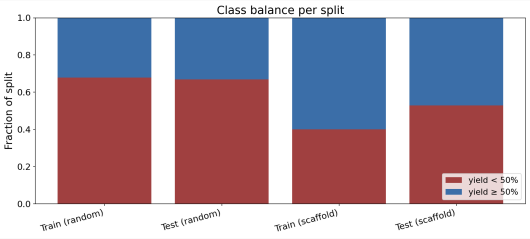
5,465 Buchwald–Hartwig amination reactions from the Open Reaction Database.

Featurisation

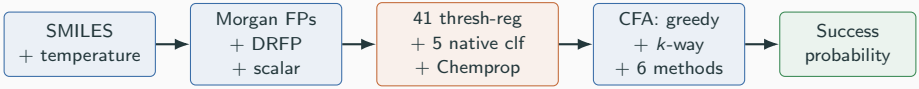
- 6 component-wise Morgan fingerprints (radius 2, 2048 bits each)
- Optional reaction-level DRFP fingerprint
- Normalised reaction temperature
- $\approx$ 14,337 features per reaction

Splits

- Random 70/15/15 (in-dist)
- Substrate-level 5-fold CV (OOD)

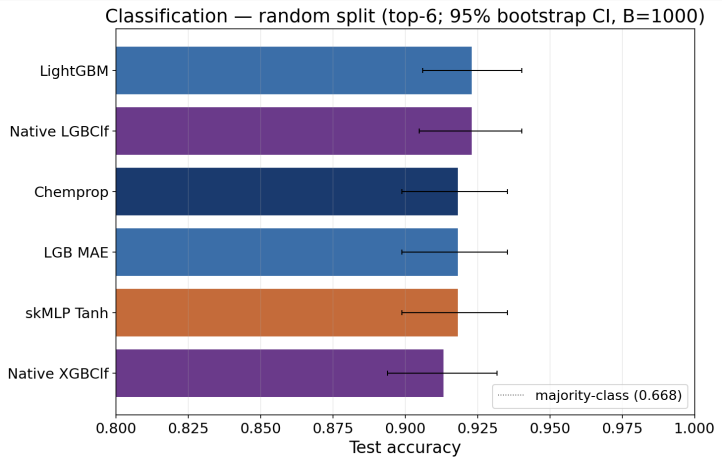


Approach: 51-model pool + CFA



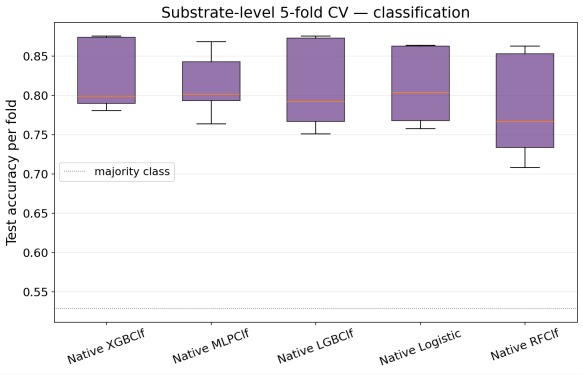
6 CFA combination methods: ASC (mean), ARC (rank-mean), WSCDS / WRCDS (diversity-weighted), WSCP / WRCP (performance-weighted). **Selection**: greedy forward + exhaustive k-way (k=2..5); val-optimal cut-off per ensemble.

Random-split results: top models with 95 % CIs



Substrate-level out-of-distribution: 5-fold CV

Each fold holds out a different set of unique aryl halides. The model never sees the test substrates during training.



Top results:

- Native_XGBClf: 0.824 ± 0.047 acc
- Native_LGBClf: 0.812 ± 0.059 acc
- CFA k-way: 0.804 ± 0.038 acc

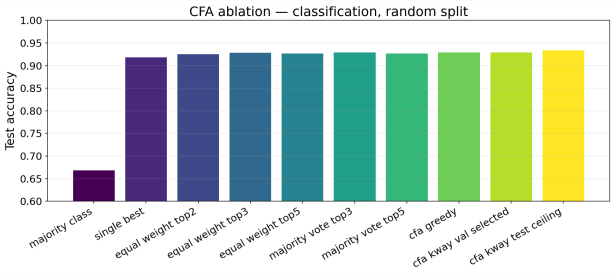
Selection stability:

- LGBClf + XGBClf chosen **5/5 folds**

Not a lucky seed.

Does CFA earn its complexity?

Comparison: CFA vs. simpler ensemble strategies on the same pool.



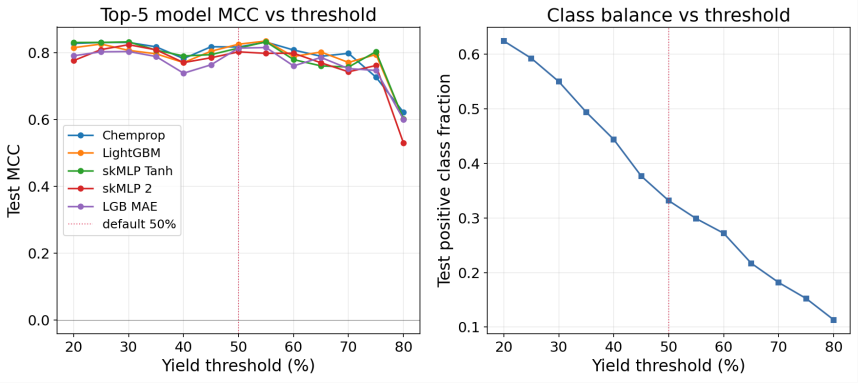
Findings:

- Random regression: CFA **+0.012 R^2** over equal-weight top-5.
- Random classification: CFA **ties** majority-vote-top-3.
- Scaffold splits: val ($n = 10$) too small to rank ensembles.

$\Longrightarrow$ CFA wins on tight regression; ties simple averaging on large-val classification.

Robustness to the success threshold

The 50 % yield threshold is from synthetic-chemistry literature, but downstream users may care about other cut-offs.



Random-split

MCC stays ≥ 0.78 across 50–75 % — the model is not overfit to the 50 % choice.

Summary & limitations

What we built

- 51-model pool + deep chemistry model (Chemprop) + CFA over 6 weighting variants.
- Two evaluation regimes: random in-distribution; substrate-level 5-fold CV for OOD.
- Statistical reliability: bootstrap 95 % CIs, multi-seed mean$\pm$std, McNemar tests, CFA ablation, threshold sweep.

92 % in-dist accuracy, 80 % OOD substrate-level accuracy.

Honest limitations

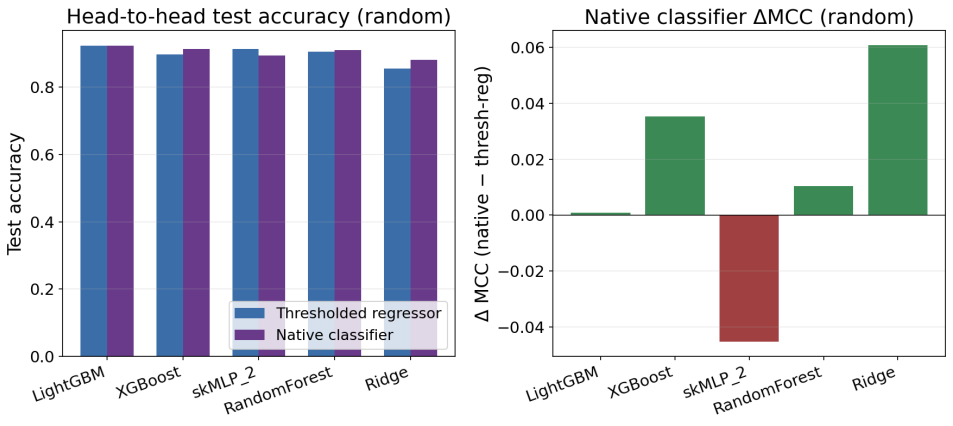
- Bemis-Murcko scaffold OOD (stricter test, extreme extrapolation): accuracy drops to 59 % with a $n = 10$ val that can't rank ensembles.
- ORD reporter bias: high-yield reactions over-represented; classifier inherits.

Multi-seed reliability

5 seeds, native classifiers, both splits.

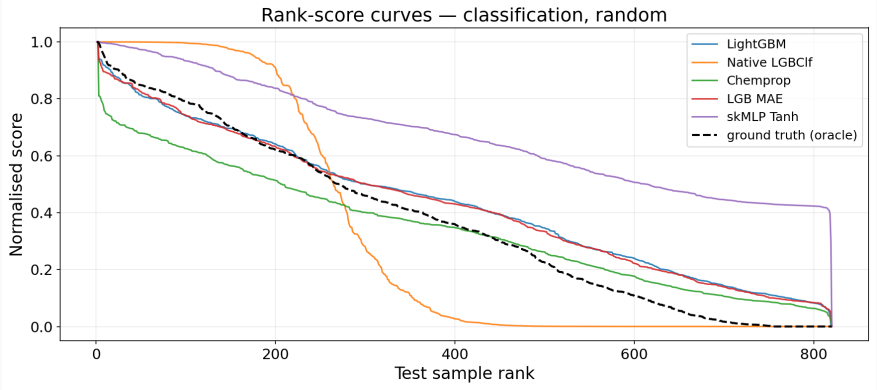
Model	Accuracy	MCC	AUC
Random split (in-dist)			
Native_LGBClf	0.910 ± 0.010	$+0.798 \pm 0.020$	0.970 ± 0.006
Native_XGBClf	0.901 ± 0.011	$+0.776 \pm 0.025$	0.966 ± 0.008
Substrate-level 5-fold CV (OOD)			
Native_XGBClf	0.824 ± 0.047	$+0.579 \pm 0.064$	0.901 ± 0.039
Native_LGBClf	0.812 ± 0.059	$+0.551 \pm 0.087$	0.897 ± 0.045

Native classifier vs thresholded regressor



Rank-score curves

CFA's signature visualisation: each model's predictions normalised to $[0, 1]$ and sorted descending; oracle = sorted ground truth.



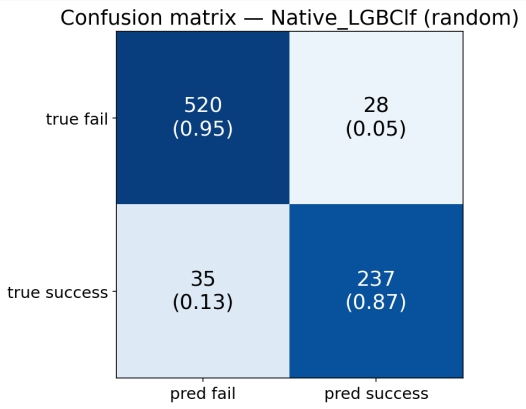
Paired statistical tests

Random classification, McNemar p-values:

A	B	A acc	B acc	McNemar p
LightGBM	Native_LGBClf	0.923	0.923	0.89 (tied)
Native_Logistic	Ridge	0.880	0.856	**0.036** (Native wins)
LightGBM	Chemprop	0.923	0.918	0.67 (tied)
Native_LGBClf	Native_XGBClf	0.923	0.913	0.29 (tied)

Scaffold-OOD highlights: Native classifiers *do* reliably beat thresholded regressors on the linear family (Logistic ¿ Ridge, $p = 0.04$); for boosted trees the two are statistically tied.

Confusion matrix

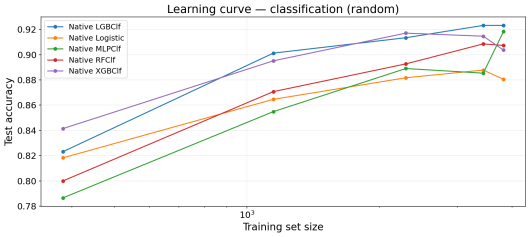


Native_LGBClf, random split. Imbalance: 67/33. Errors are roughly balanced — not failing on minority class.

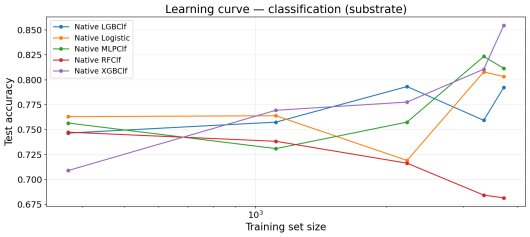
Learning curves — "would more data help?"

Subsample train at 10/30/60/90/100 %; retrain; evaluate on held-out test.

In-distribution (saturating)



Substrate-OOD (still rising)



In-distribution curves plateau by 3,800 rows; substrate-level OOD curves are *still rising* at 100 % of available data — empirical evidence that more chemistry data would lift OOD accuracy.